

Homework #1: 2025학년 1학기

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주어진 코드를 수정하여 수행한 모폴로지 연산의 예를 2개 이상 작성

<Dilation 팽창>

```
# 파이썬 뉴턴리터 함수 정의
import matplotlib.pyplot as plt
import numpy as np

def plot_image(image, title=''):
    plt.imshow(image)
    plt.axis('off')
    plt.title(title, size=20)

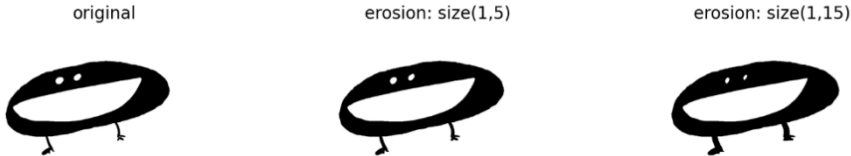
#1. 팽창
from skimage.morphology import binary_erosion, binary_dilation, binary_opening, binary_closing, rectangle, disk, square
import matplotlib.pyplot as plt
from skimage import io, color

im = color.rgb2gray(io.imread('/content/drive/MyDrive/task/어깨걸이극락조.jpg'))
im[im <= 0.5] = 0 # 이진 영상 만들기
im[im > 0.5] = 1

im1 = binary_erosion(im, rectangle(1,5))
im2 = binary_erosion(im, rectangle(1,15))

plt.figure(figsize=(20,10))
plt.gray()
plt.subplot(131), plot_image(im, 'original')
plt.subplot(132), plot_image(im1, 'erosion: size(1,5)')
plt.subplot(133), plot_image(im2, 'erosion: size(1,15)')
```

<python-input-47-09da9e38a7c7>:18: FutureWarning: 'rectangle' is deprecated since version 0.25 and will be removed in version 0.27. Use 'skimage.morphology.
im1 = binary_erosion(im, rectangle(1,5))
<python-input-47-09da9e38a7c7>:19: FutureWarning: 'rectangle' is deprecated since version 0.25 and will be removed in version 0.27. Use 'skimage.morphology.
im2 = binary_erosion(im, rectangle(1,15))
(Axes: title='center': 'erosion: size(1,15)')>, None)



<Erosion 침식>

```
# 파이썬 뉴턴리터 함수 정의
import matplotlib.pyplot as plt
import numpy as np

def plot_image(image, title=''):
    plt.imshow(image)
    plt.axis('off')
    plt.title(title, size=20)

#2. 침식
from skimage.morphology import binary_erosion, binary_dilation, binary_opening, binary_closing, rectangle, disk, square
import matplotlib.pyplot as plt
from skimage import io, color

im = color.rgb2gray(io.imread('/content/drive/MyDrive/task/어깨걸이극락조.jpg'))
im[im <= 0.5] = 0
im[im > 0.5] = 1

im1 = binary_dilation(im, disk(2))
im2 = binary_dilation(im, disk(4))

plt.figure(figsize=(20,10))
plt.gray()
plt.subplot(131), plot_image(im, 'original')
plt.subplot(132), plot_image(im1, 'dilation: disk(2)')
plt.subplot(133), plot_image(im2, 'dilation: disk(4)')
```



문제 발생시 자신 만의 해결 방법 서술

교수님께서 올려 주신 코드를 통해 실습을 하다 보니 원본 이미지에 모폴로지 연산을 밑의 순서대로 하면 뭐가 달라질 지 궁금해졌다.

1. **Erosion 후 Dilation:** 작은 노이즈가 없어졌다가 다시 원래 모양으로 돌아오는지 확인했다.
2. **Closing 후 Opening:** 작은 구멍을 없애고 테두리를 정리하면 어떻게 변하는지 실험해봤다.

<Opening: 열림(침식+팽창) 닫힘(팽창+침식)>

```
# 파이썬 유틸리티 함수 정의
import matplotlib.pyplot as plt
import numpy as np

def plot_image(image, title=''):
    plt.imshow(image)
    plt.axis('off')
    plt.title(title, size=20)

# 3. Opening: 열림(침식+팽창) 닫힘(팽창+침식)
from skimage.morphology import binary_erosion, binary_dilation, binary_opening, binary_closing, rectangle, disk, square
import matplotlib.pyplot as plt
from skimage import io, color

im = color.rgb2gray(io.imread('/content/drive/MyDrive/task/어깨걸이극락조.jpg'))
im[im <= 0.5] = 0
im[im > 0.5] = 1

im1 = binary_opening(im, disk(12))
im2 = binary_closing(im, disk(6))

plt.figure(figsize=(20,10))
plt.gray()
plt.subplot(1,3,1), plot_image(im, 'original')
plt.subplot(1,3,2), plot_image(im1, 'opening disk size 12')
plt.subplot(1,3,3), plot_image(im2, 'closing disk size 6')
plt.show()
```

original



opening disk size 12



closing disk size 6



1: Erosion (침식) 후 Dilation (팽창)

2: Closing 닫힘(팽창+침식) 후 Opening 열림(침식+팽창)

```
import matplotlib.pyplot as plt
import numpy as np
from skimage.morphology import binary_erosion, binary_dilation, binary_opening, binary_closing, rectangle, disk
from skimage import io, color

def plot_image(image, title=''):
    plt.imshow(image, cmap='gray')
    plt.axis('off')
    plt.title(title, size=20)

im = color.rgb2gray(io.imread('/content/drive/MyDrive/task/어깨걸이극락조.jpg'))
im[im <= 0.5] = 0 # 이진 영상 만들기
im[im > 0.5] = 1

# 1: Erosion (침식) 후 Dilation (팽창)
im_eroded = binary_erosion(im, disk(5))
im_dilated = binary_dilation(im_eroded, disk(5))

# 2: Closing 닫힘(팽창+침식) 후 Opening 열림(침식+팽창)
im_closed = binary_closing(im, disk(8))
im_opened = binary_opening(im_closed, disk(8))

plt.figure(figsize=(20, 10))
plt.subplot(1,5,1), plot_image(im, 'Original')
plt.subplot(1,5,2), plot_image(im_eroded, 'Erosion (disk 5)')
plt.subplot(1,5,3), plot_image(im_dilated, 'Erosion → Dilation')
plt.subplot(1,5,4), plot_image(im_closed, 'Closing (disk 8)')
plt.subplot(1,5,5), plot_image(im_opened, 'Closing → Opening')
plt.show()
```

Original



Erosion (disk 5)



Erosion → Dilation



Closing (disk 8)



Closing → Opening



▣ 문제 발생시 자신만의 해결 방법 서술

이를 통해 알게 된 점

모폴로지 연산을 사용해 보면서, 이미지에서 작은 노이즈나 구멍을 없애는 데 정말 유용하다는 걸 알게 되었다.

1. **Erosion 후 Dilation**을 하면서 노이즈가 제거되고, 원래의 형태가 복원되는 과정을 눈으로 확인할 수 있었다.

2. **Closing 후 Opening**을 해보니, 이미지의 작은 구멍들이 사라지고 윤곽선이 정리되는 것을 볼 수 있었다.

이 과정들을 통해, 모폴로지 연산이 이미지의 품질을 개선하는 데 어떻게 도움을 줄 수 있는지 더 잘 이해하게 되었다.

<소스코드>

```
import matplotlib.pyplot as plt
import numpy as np
from skimage.morphology import binary_erosion, binary_dilation, binary_opening,
binary_closing, rectangle, disk
from skimage import io, color

def plot_image(image, title=''):
    plt.imshow(image, cmap='gray')
    plt.axis('off')
    plt.title(title, size=20)

im = color.rgb2gray(io.imread('/content/drive/MyDrive/task/어깨걸이극락조.jpg'))
im[im <= 0.5] = 0 # 이진 영상 만들기
im[im > 0.5] = 1

# 1: Erosion (침식) 후 Dilation (팽창)
im_eroded = binary_erosion(im, disk(5))
im_dilated = binary_dilation(im_eroded, disk(5))

# 2: Closing 닫힘(팽창+침식) 후 Opening 열림(침식+팽창)
im_closed = binary_closing(im, disk(8))
im_opened = binary_opening(im_closed, disk(8))

plt.figure(figsize=(20, 10))
plt.subplot(151), plot_image(im, 'Original')
plt.subplot(152), plot_image(im_eroded, 'Erosion (disk 5)')
plt.subplot(153), plot_image(im_dilated, 'Erosion → Dilation')
plt.subplot(154), plot_image(im_closed, 'Closing (disk 8)')
plt.subplot(155), plot_image(im_opened, 'Closing → Opening')
plt.show()
```

(캡처 예) : 코드와 실행화면이 같이 나오도록 캡처

```
# Erosion(침식)
from skimage.morphology import binary_erosion, binary_dilation, binary_opening, binary_closing, rectangle, disk, square
import matplotlib.pyplot as plt
from skimage import io, color

im = color.rgb2gray(io.imread('clock2.jpg'))
im[im <= 0.5] = 0 # 이진 영상 만들기
im[im > 0.5] = 1

im1 = binary_erosion(im, rectangle(1,5))
im2 = binary_erosion(im, rectangle(1,15))

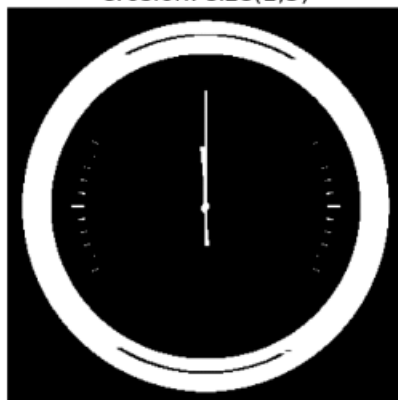
plt.figure(figsize=(20,10))
plt.gray()
plt.subplot(131), plot_image(im, 'original')
plt.subplot(132), plot_image(im1, 'erosion: size(1,5)')
plt.subplot(133), plot_image(im2, 'erosion: size(1,15)')
plt.show()
```

<ipython-input-2-1095c235bc25>:10: FutureWarning: 'rectangle' is deprecated since version 0.25 and will be removed in version 0.27. Use 'skimage.morphology.footprint.rectangle' instead.
im1 = binary_erosion(im, rectangle(1,5))
<ipython-input-2-1095c235bc25>:11: FutureWarning: 'rectangle' is deprecated since version 0.25 and will be removed in version 0.27. Use 'skimage.morphology.footprint.rectangle' instead.
im2 = binary_erosion(im, rectangle(1,15))

original



erosion: size(1,5)



erosion: size(1,15)

